LINUS LINUS

Senior Full-Stack, Systems & Desktop Software Architect

📍 West Africa (WAT) | Open to Worldwide Remote
📞 +234 812 048 4386 | +234 811 366 4697
✉️ linus.linus204@gmail.com | linus.zoea@gmail.com
🌐 GitHub: @Learnuel1 | LinkedIn: linus-linus-i

EXECUTIVE SUMMARY

Accomplished Senior Full-Stack & Systems Software Architect with proven expertise delivering enterprise Windows desktop financial accounting suites, real-time on-demand logistics marketplaces, government loan management engines, and secure cloud collaboration platforms. Proficient across C#/.NET Framework, Node.js, Express, React 18, Microsoft SQL Server (90+ T-SQL Stored Procedures), MongoDB, Socket.io, and double-entry financial ledgers. Deep background in asynchronous multi-threading, concurrency prevention locks, regulatory KYC (BVN, NIN, CAC), hardware-bound anti-tamper licensing, and defensive API security.

CORE TECHNICAL COMPETENCIES

- **Programming Languages:** C# (5.0/6.0), JavaScript (ES6+, ESNext, Node.js v18/v20), TypeScript, T-SQL, HTML5, CSS3, Handlebars, EJS
- **Desktop & Systems Engineering (.NET):** .NET Framework 4.5, Windows Forms, ADO.NET (`SqlCommand`, `SqlDataAdapter`, `SqlHelper`), WMI Hardware Fingerprinting (SHA-256), Multi-Threaded Asynchronous Engines (`Task.Run()`), Windows Isolated Storage, Factory Pattern, LiveCharts (WinForms/WPF), Bunifu UI, MetroFramework
- **Reporting & Barcode Engines:** SAP Crystal Reports 13.x, Zen Barcode Rendering Framework (Code128 1D & QR Code 2D), `iTextSharp` PDF generation, `ExcelDataReader` (binary `.xlsx`/`.xls` parsing)
- **Backend & Distributed Systems:** Node.js, Express.js 4.x, Socket.io (WebSocket real-time hub), Node.js `EventEmitter`, Dual-Route Gateway Architecture (`/api/v1` vs `/api/v1/m`), Layered Service Architecture, RESTful APIs, Swagger / OpenAPI 2.0, Zod Runtime Validation

- **Databases & Persistence:** Microsoft SQL Server (2012–2022 / Express), 90+ Stored Procedures, Triggers, Views, ACID Transactions, File-level database backup/restore (`.mdf`/`.ldf`), MongoDB & Mongoose 6.x (Aggregation pipelines, soft deletes, indexing)
- **FinTech & Financial Ledgers:** Double-entry wallet ledgers (`Wallet` & `WalletHistory` / `Transactions`), Platform escrow settlement, Commission calculation engines, Paystack API & Transfer API, Flutterwave SDK, Bloc API, Bank Verification Number (BVN) resolution, NIP commercial bank lookup, Bank teller slip reconciliation
- **Geospatial & Logistics:** Google Maps Platform (Geocoding & Distance Matrix API), Haversine geo-radius courier dispatching (`getOrdersWithinRadius`), Courier GPS & Geofencing, Cryptographic QR Code Handover
- **Security & Cryptography:** WMI hardware fingerprinting (SHA-256 machine ID), JWT symmetric claims obfuscation (Base64/Hex), Two-Factor Authentication (2FA), `bcryptjs`, AES/TripleDES connection string encryption, RBAC multi-tier guards
- **Cloud Infrastructure & Secrets:** AWS Secrets Manager, AWS S3, Cloudinary SDK, Multer streaming, Render Cloud Platform, Git/GitHub
- **Frontend Frameworks & State:** React 18, Vite, Next.js (App Router), Redux Toolkit (`@reduxjs/toolkit`), RTK Query (auto 401 token refresh), `redux-persist`, MobX, React Router DOM v6, Tailwind CSS, `@cyntler/react-doc-viewer`, `react-file-viewer`

PROFESSIONAL EXPERIENCE & FEATURED SYSTEMS

Lead Desktop & Systems Architect — Cash Desk Application

Enterprise .NET / C# Desktop Bursary, POS & Campus Commerce Accounting Suite

- Architected an enterprise-grade Windows desktop financial management and accounting suite tailored for primary and secondary educational institutions.
- **Asynchronous Multi-Threaded Debt Engine (`GroundWorker`):** Designed non-blocking background workers using `Task.Run()` to execute concurrent term debt audits (`FirstTermDebtAsync`, `SecondTermDebtAsync`, `ThirdTermDebtAsync`), zero-payment scans, and duplicate debtor deduplication on application boot without freezing the UI.
- **Student Digital Wallet ("Pocket Money"):** Engineered a cashless payment ecosystem tied 1:1 to student matriculation numbers, supporting individual deposits, bulk parent funding by class/arm (`BulkFund`), and automated Balance Brought Forward (`BF`) session rollover.

- **Point of Sale (POS) & Store Inventory Subsystem:** Built high-speed barcode checkout supporting Code128 1D and QR 2D scanning (`Zen.Barcode`), multi-tender settlements (Pocket Money, Cash, Bank, Issued, Credit), and closing stock term reconciliation.
- **Hardware-Bound Anti-Tamper Licensing:** Created a machine licensing engine querying WMI (`Win32_Processor.ProcessorId` and `Win32_BIOS.SerialNumber`) to compute a machine SHA-256 hash, verified against activation keys stored across two-factor storage (SQL database + Windows Isolated Storage) with a 120-record demo trial safeguard.
- **Database Architecture:** Developed 90+ compiled SQL Server stored procedures enforcing ACID transactional integrity and direct file-level `.mdf`/`.ldf` backup/restore scripts.
- **Reporting Engine:** Integrated SAP Crystal Reports (15+ `.rpt` templates) and thermal POS receipt printing (58mm/80mm).

Lead Distributed Systems & Backend Architect — Grubbex Backend API

Multi-Sided On-Demand Food & Grocery Logistics Marketplace

- Engineered high-concurrency backend API and real-time WebSocket hub connecting Shoppers, Couriers (Riders), Merchant Stores, and Platform Administrators.
- **Dual-Route Platform Isolation:** Architected dual-entry routing separating Web clients (`/api/v1`) from Mobile clients (`/api/v1/m`), strictly enforced via `checkRouteUsed` to prevent unauthorized client impersonation.
- **Real-Time Geo-Radius Courier Dispatching:** Built proximity matching using Google Maps Geocoding and Haversine distance calculations (`getOrdersWithinRadius`) to broadcast pending store pickups to couriers within active geo-fences via Socket.io.
- **Cryptographic Delivery Handover:** Implemented fraud-proof delivery handovers pairing dynamic QR code generation (`qrcodeService`) with 6-digit cryptographic PINs verified by couriers on customer handover.
- **Double-Entry Financial Ledger & Escrow:** Developed an automated financial settlement engine where customer payments are held in platform escrow, courier commissions are programmatically deducted upon verified handover, and rider/merchant wallets are credited with full audit trails in `WalletHistory`.
- **Validation & Cloud Security:** Implemented centralized Zod runtime schema pipelines and secured production credentials via AWS Secrets Manager.

Lead Backend Architect & API Engineer — Bayelsa State Higher Education Loans Board (BYSHELB LMS Backend v2)

Distributed Government Loan Processing Engine & Administrative Governance Platform

- Architected the core central backend powering tertiary student loans for Bayelsa State, managing student applications, KYC validation, civil service governance, and multi-million naira loan disbursements.
- **Multi-Tier Administrative Governance Pipeline:** Built an 8-role permission hierarchy (`super_admin`/`es`, `lph`, `la`, `lo`, `acc`, `fdo`, `blog`, `student`) enforcing strict state transitions.
- **Internal Approval Memo Routing:** Developed an internal bureaucratic memo system where verified student loans are bundled into cryptographically tracked memos flowing across audit, legal, and executive sign-off before accountant disbursement.
- **Active Concurrency Prevention Lock:** Implemented a database-level lock mechanism (`LoanProfileView`) preventing multiple loan officers from making conflicting assessments on the same application simultaneously.
- **KYC & Banking Automation:** Integrated live Bank Verification Number (BVN) resolution and NIP commercial bank account lookup to prevent identity fraud and ensure disbursement accuracy.
- **Security & Obfuscation:** Designed custom Base64/Hex symmetric claim encryption inside JWT payloads to prevent token inspection on client devices.

Lead Frontend Engineer — BYSHELB Student Loan Portal Frontend (`lms-fe-v2`)

Citizen-Facing Tertiary Loan Application Platform & Multi-Step Wizard Engine

- Built an accessible, responsive student loan portal serving thousands of tertiary students across Bayelsa State.
- **3-Tier Onboarding Gatekeeper:** Enforces bio-data, verified emergency contact, and bank account resolution before unlocking loan applications.
- **7-Stage Progressive Application Wizard:** Architected an isolated step provider (`UndergraduateLoanStepProvider`) managing stage persistence, form state, and draft resumption across undergraduate, postgraduate, and professional loan tracks.
- **Payment & Document Workflows:** Integrated Flutterwave inline payment processing with automated server-side verification callbacks, official guarantor PDF download generation, and document attestation uploads.

Full-Stack Web Application Engineer — SecureSync Cloud Storage Platform (`secure-sync-js`)

Modern Encrypted Cloud Storage & Document Collaboration Platform

- Architected a full-featured cloud document management web application with granular sharing and multi-factor authentication.

- **RTK Query Auto Token Rotation:** Engineered an API data-fetching layer with automatic JWT bearer attachment, silent token re-authentication on 401 Unauthorized responses, and optimistic state caching.
- **In-App Multi-Format Document Viewer:** Native in-browser rendering for PDF, Word (`.docx`), Excel (`.xlsx`), PowerPoint (`.pptx`), and imagery using `@cyntler/react-doc-viewer`.
- **Granular Sharing & Access Control:** Built sharing modals allowing users to grant View-only or Download permissions, generate public reference preview links, and revoke access instantly.
- **Security & Quota Engine:** 2FA challenge flow with OTP countdown timers, soft-deletion with trash restoration, and dynamic storage quota meter.

EDUCATION & PROFESSIONAL DEVELOPMENT

- **Bachelor of Science (B.Sc.) in Computer Science / Software Engineering**
- **Continuous Technical Mastery:** Real-Time Distributed Systems, Financial Ledger Architecture, Cloud Security & Cryptography

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NOTABLE OPEN SOURCE & GITHUB REPOSITORIES

- **GitHub Profile:** https://www.github.com/Learnuel1
- **Portfolio Repository:** https://github.com/Learnuel1/linus-linus-portfolio
- **Flagship Codebases:** Cash Desk Application, Grubbex Logistics API, BYSHELB LMS Backend v2, SecureSync Cloud Storage